

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		15392	
Total Threats Mitigated	16	Total Packets Analyzed		7497455061	
High Priority	7	Malicious Packets		7478600530	
Low Priority	9	True Negatives Passed		10879	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		0	
False Positives	3	UDP Flood		13	
FP Rate	0.03%	Uncertain		3	
IF Precision	99.93%				
IF Recall	96.09%	Action		Count	
IF F1-Score	97.97%	Quarantined		3	
IF Accuracy	98.80%	Time Ban		3	
		Blackhole		0	
		Released		1	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	99.93%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	96.09%	Share of actual attacks successfully flagged
F1-Score	97.97%	Balanced measure combining Precision and Recall
Accuracy	98.80%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.03%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	3.91%	Actual attacks that went undetected
True Positive Rate (TPR)	96.09%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.97%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4472	FN 182
Actual: Normal	FP 3	TN 10697

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	99.93%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	99.97%	Balanced measure combining Precision and Recall
Accuracy	99.93%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	100.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	0.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	4472

2c. Hybrid Cascaded

Metric	Value	Description
Precision	99.93%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	96.09%	Share of actual attacks successfully flagged
F1-Score	97.97%	Balanced measure combining Precision and Recall
Accuracy	98.80%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.03%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	3.91%	Actual attacks that went undetected
True Positive Rate (TPR)	96.09%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	99.97%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4472	FN 182
Actual: Normal	FP 3	TN 10697

2d. Response Latency

Metric	Value	Description
Detection Time	436.4 ms	Average time taken to detect an attack
Mitigation Response Time	3.3 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	0.89%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	6.37%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	4.79%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.20	4	3	2026-09-26 21:48:21	2026-09-26 21:51:01
10.0.0.21	4	3	2026-09-26 21:48:26	2026-09-26 21:51:06
10.0.0.22	4	3	2026-09-26 21:48:22	2026-09-26 21:51:02
10.0.0.9	1	1	2026-09-26 21:47:44	2026-09-26 21:48:14

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 21:47:44	10.0.0.9	DDoS	UDP Flood	85.3%	High	Time Ban (30s)
2026-09-26 21:48:14	10.0.0.9	DDoS	UDP Flood	85.3%	High	Released
2026-09-26 21:48:21	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 21:48:22	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 21:48:26	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 21:48:31	10.0.0.20	DDoS	UDP Flood	76.7%	High	Time Ban (30s)
2026-09-26 21:48:32	10.0.0.20	DDoS	UDP Flood	76.7%	High	Time Ban
2026-09-26 21:48:32	10.0.0.22	DDoS	UDP Flood	86.0%	High	Time Ban (30s)
2026-09-26 21:48:33	10.0.0.22	DDoS	UDP Flood	86.0%	Critical	Time Ban
2026-09-26 21:48:36	10.0.0.21	DDoS	UDP Flood	82.3%	High	Time Ban (30s)
2026-09-26 21:48:37	10.0.0.21	DDoS	UDP Flood	82.3%	Critical	Time Ban
2026-09-26 21:49:32	10.0.0.20	DDoS	UDP Flood	77.5%	Critical	Time Ban (1m)
2026-09-26 21:49:33	10.0.0.22	DDoS	UDP Flood	86.0%	High	Time Ban (1m)
2026-09-26 21:49:37	10.0.0.21	DDoS	UDP Flood	84.5%	Critical	Time Ban (1m)
2026-09-26 21:51:02	10.0.0.22	DDoS	UDP Flood	77.3%	Critical	Time Ban (2m)
2026-09-26 21:51:06	10.0.0.21	DDoS	UDP Flood	77.7%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.6461	77.7%	Critical	Phase 2	1	10s	2026-09-26 21:51:06	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6269	77.3%	Critical	Phase 2	1	10s	2026-09-26 21:51:02	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6479	76.2%	Critical	Phase 2	1	10s	2026-09-26 21:51:01	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6457	84.5%	Critical	Phase 2	3	1m 11s	2026-09-26 21:49:37	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6305	86.0%	High	Phase 2	3	1m 11s	2026-09-26 21:49:33	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6312	77.5%	Critical	Phase 2	3	1m 11s	2026-09-26 21:49:32	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6463	84.5%	High	Phase 2	2	41s	2026-09-26 21:49:07	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6345	86.0%	High	Phase 2	2	41s	2026-09-26 21:49:03	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6007	77.1%	High	Phase 2	2	41s	2026-09-26 21:49:02	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6458	82.3%	High	Phase 2	1	10s	2026-09-26 21:48:36	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6352	86.0%	High	Phase 2	1	10s	2026-09-26 21:48:32	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6383	76.7%	High	Phase 2	1	10s	2026-09-26 21:48:31	Attack persisted - escalating to time ban
10.0.0.9	UDP Flood	0.6100	85.3%	High	Phase 2	1	30s	2026-09-26 21:48:14	Ban Expired

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE